Every Heart Is Different: Patient Context for Rare-Arrhythmia Detection in New Patients

A pre-registered study of generative augmentation, with exploratory patient-relative pipelines on the MIT-BIH inter-patient benchmark

CS787: GenAI, course project report (10-page version)

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Abstract

Beat classifiers trained on the MIT-BIH Arrhythmia Database detect normal and ventricular beats well, but in patients absent from training they miss most supraventricular (S) and fusion (F) beats, the rare classes with clinical weight. Class-conditional generative models promise to supply the missing rare beats. We tested that promise in a pre-registered study under the inter-patient protocol: selection rules fixed before the experiments, every choice made on validation patients, and the 22 test patients read once. Across two classifiers and a 96-run grid, the conditional VAE and WGAN-GP we trained did not detectably improve the rare-class macro-F1 over no augmentation (best difference +0.025, 95% CI −0.018 to +0.047) or over class weighting, random oversampling and SMOTE, and none of our synthetic-quality metrics predicted the gain: the synthetic beats reproduced the few training patients they were learned from. Twenty-three follow-up analyses, each registered before it ran but all read on the same test patients, are therefore exploratory. They point to patient context: expressing each beat relative to the patient's own beats, judging prematurity against the patient's rhythm, vetoing S calls during atrial fibrillation and averaging ten runs raised the test score from 0.113 to 0.587. On 977 patients that no analysis had used, the frozen pipeline reached an S F1 of 0.294 with its original thresholds, and 0.414 with thresholds adapted on 388 separate Icentia11k patients; ten-run averaging and patient-chosen thresholds were confirmed there, while the prematurity rule was not. Evaluated on already-seen patients, the same network would have scored 0.837 instead of 0.088.

1 Introduction

Ambulatory monitors record the ECG for one or more days [21]; at 70 beats per minute a day holds about 100,000 beats, far more than anyone reads. Software therefore classifies every beat and flags the rare ones: supraventricular ectopic (S) beats, whose burden predicts atrial fibrillation and stroke [3]; ventricular ectopic (V) beats, whose burden goes with a weakened left ventricle [2]; and fusion (F) beats. The standard benchmark is the MIT-BIH Arrhythmia Database [14] with the beat classes of the AAMI standard [1].

The hard part is the patient. When the test patients are kept out of training, as de Chazal et al.'s inter-patient protocol [5] requires and as a clinic faces, published methods detect S beats with an F1 of 0.511 to 0.686 [5, 10, 24], and F beats, where reported, with 0.477 [24]. Many studies instead split beats at random, so that every patient appears in both training and test, and report far higher scores [11]; those scores largely measure how well a model recognises patients it has already seen.

Rare beats are scarce and concentrated. The training split holds 644 S and 388 F beats against 35,496 normal ones, and in the test split one patient carries 1,381 of the 1,836 S beats and another 362 of the 388 F beats. Class-conditional generative models, conditional VAEs [20] and Wasserstein GANs [8], can synthesise convincing rare beats, and augmentation with GANs and diffusion models has been reported to improve MIT-BIH classifiers [18, 11], mostly on random splits. We asked whether the remedy survives the patient boundary, and registered the question before reading any test data:

> *Under the standard DS1/DS2 inter-patient protocol, does class-conditional generative augmentation improve detection of the rare classes S and F, compared with no augmentation, a class-weighted loss, random oversampling and SMOTE?*

The protocol's one known overlap, records 201 and 202 from the same person, is identified in Section 3 and handled by a sensitivity analysis. Every rule that chose a model, a variant or a strength was fixed in advance and applied to validation patients at a gate closing each phase; the test file opened once, for the final evaluation. In all, 761 runs are logged. The report makes four contributions:

1. **A pre-registered negative result** for the generators, data and selection policies we tested: they did not detectably improve the detection of rare arrhythmias in unseen patients over three conventional remedies, and our synthetic-quality metrics did not predict their usefulness; a nearest-patient analysis explains why (Sections 5.1 and 5.2).
2. **An exploratory patient-context pipeline** that raises the rare-class macro-F1 on the inter-patient test patients from 0.113 to 0.587 (Section 5.3), with a step-by-step account of what each change contributed and an independent check of the frozen pipeline on 977 untouched patients (Section 5.4).
3. **A same-pipeline measurement of patient overlap**: the rare-class macro-F1 rises from 0.088 to 0.837 when the test patients were seen in training, consistently over three random splits (Section 5.5).
4. **An explainable system**: per-segment saliency and notes written by a language model from the classifier's output alone, under a fact checker that rejects any claim not in that output (Section 5.6).

2 Related work

Evaluation. The AAMI EC57 standard [1] groups the MIT-BIH labels into the classes N, S, V, F and Q and asks for record-by-record reporting; de Chazal et al. [5] split the records into DS1 for training and DS2 for testing so that no patient is in both. Studies that split beats at random report far higher scores [11]; we measure the gap between the two protocols on one pipeline.

Rhythm and patient context. The strongest inter-patient methods rely on RR intervals measured against the patient's own rhythm: Lin and Yang [12] divide them by the recording's mean, Huang et al. [10] call a beat S when it comes early against that mean, and Zhu et al. [24] feed RR intervals to a convolutional network. Our patient-relative beat extends the same idea from timing to morphology.

Generative augmentation of ECG. GANs [18] and conditional diffusion models [11] have been used to augment MIT-BIH classifiers, and SimGANs [6] constrain a GAN with McSharry et al.'s ECG simulator [13]. We compare conditional VAEs [20] and WGAN-GPs [8] with SMOTE [4], random oversampling and a class-weighted loss under the inter-patient protocol, and test diffusion and the simulator loss as follow-ups.

Explanations. We compare Grad-CAM [17], integrated gradients [22], SmoothGrad [19] and occlusion by faithfulness, and restrict language-model notes to the classifier's output by a fact checker.

3 Problem setup and protocol

Data. The MIT-BIH Arrhythmia Database [14, 7] holds 48 half-hour two-lead recordings whose beats were labelled by cardiologists. We use lead MLII at 360 Hz, map the labels to the AAMI classes N, S, V and F, and exclude the four paced records and the few Q beats. The 22 DS1 records give training (17) and validation (5) patients and the 22 DS2 records the test patients (Table 1); generators, oversamplers and normalisation see training patients only. An S beat comes early but looks almost like the patient's normal beat, a V beat is wide and different, and an F beat lies between the two.

Split	Records	N	S	V	F
training	17	35,496	644	3,030	388
validation	5	10,370	300	758	27
test (DS2)	22	44,259	1,837	3,221	388

Table 1: Records and beats per split. S beats are under 4% of each split, and F beats at most 1%.

Measure. For class c with sensitivity Se_c and positive predictivity PPV_c, $\mathrm{F1}_c = 2\,\mathrm{Se}_c\mathrm{PPV}_c/(\mathrm{Se}_c + \mathrm{PPV}_c)$. The primary measure is the rare-class macro-F1 on the test patients,

$$M = \tfrac{1}{2}\left(\mathrm{F1}_S + \mathrm{F1}_F\right), \tag{1}$$

with 95% confidence intervals from 10,000 bootstrap resamples of test records (patients), paired differences between conditions, and Holm's correction [9] within each family of comparisons; the confirmatory family holds the four comparisons of a generative condition with no augmentation.

Pre-registration and test lock. Every selection rule and hypothesis was written down before the experiment it governs, and each phase ended with a gate at which the rule was applied and approved. The test array can only be read through a guarded loader that the final evaluation alone unlocks, and every read is logged. The 23 follow-up analyses (P1–P23) were each registered before they ran, in versions 4 to 21 of the pre-registration, are reported apart from the primary results, and are labelled post hoc.

Three levels of evidence. Every result below carries one of three labels. *Confirmatory*: the pre-registered comparison, decided on validation patients and read once on the test patients (Section 5.1). *Exploratory*: the follow-up analyses, each registered before it ran, but all read on the same 22 test patients after the final evaluation, so their test scores are subject to adaptive overfitting and are not unbiased estimates (Section 5.3). *Independent*: the frozen follow-up pipeline on 977 Icentia11k patients that no analysis had used, with nothing fitted on them (Section 5.4); validation-patient results, which never involve the test set, are also reported where they exist.

Patient identity. MIT-BIH has 48 records from 47 people: records 201 and 202 are the same person. Record 201 is a validation record, so no classifier was trained on that person, but it took part in model selection while 202 is a test record; every test result is also reported without record 202, and the conclusions do not change. In the follow-up analyses, INCART patients were identified from the record headers and split by patient; SVDB records carry no patient identity and were treated as patients, which is an assumption, not a proof, since two SVDB records could come from one person; the 400 Icentia11k training patients and the 977 evaluation patients were drawn disjointly, and DS2 was never used for training.

4 Methods

4.1 The pre-registered pipeline

Each stage was chosen on validation patients by a registered rule (Figure 1a, left column). **Beat.** A 0.5 s window around the reference R peak (180 samples), z-scored per beat, with three RR features as side input: the intervals before and after the beat and the ratio of the former to the mean of the previous ten (the best of four representations on validation S F1: 0.296, against 0.164 without the RR features, 0.001 for a 1 s band-passed window and 0.035 at 125 Hz). **Classifiers.** A shallow CNN (73k parameters) and a BiLSTM (135k), the best of four architectures, trained with three seeds and early stopping on validation S F1 (validation holds too few F beats to guide selection); pretraining on PTB-XL did not help. **Generators.** A conditional VAE and a WGAN-GP, each the best of four designs in a pilot, trained on training patients only with three seeds into frozen banks of synthetic beats. **Augmentation.** Each rare class c with n_c real training beats is brought up to a fraction $s \in \{0.1, 0.25, 0.5\}$ of the normal count n_N with $m_c = s\,n_N - n_c$ synthetic beats, taken from $2m_c$ bank beats as the m_c closest to real training beats by dynamic time warping [16]; s is chosen per condition on validation. With no augmentation, a class-weighted loss, random oversampling and SMOTE as competitors, this is a grid of 96 runs (2 classifiers, 3 seeds, no augmentation and 5 conditions at 3 strengths). The two classifiers are the best convolutional and the best recurrent design, so that every conclusion is checked on two architecture families; the DTW screen won the policy pilot on validation against unscreened policies. **Training.** Cross-entropy (class-weighted only in that condition), AdamW with weight decay 0.01, batches of 256, a learning rate of 10^{-3} or $3 \cdot 10^{-4}$ chosen on validation, at most 40 epochs, and early stopping with patience 5 on validation S F1, never before epoch 10; every condition has the same budget.

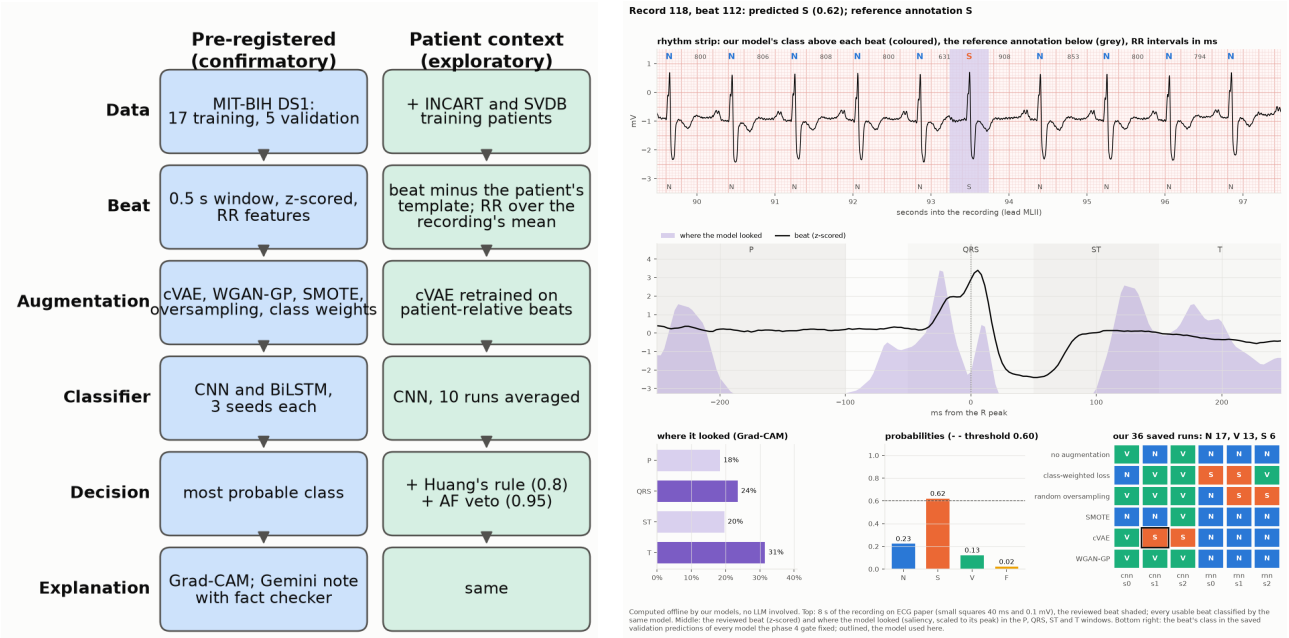
4.2 The patient-context pipeline

The follow-up analyses replaced population-level inputs with patient-relative ones (Figure 1a, right column). For beat b with z-scored window x_b and R peak at time τ_b, the patient template is the element-wise median of the windows of every beat of the same recording within 150 s, and the classifier sees the deviation from it:

$$\tilde{x}_b = x_b - t_b, \qquad t_b = \operatorname*{median}_{j:\,|\tau_j - \tau_b| \le 150\,\mathrm{s}} x_j. \tag{2}$$

Figure 2 shows the effect: the shape of a normal beat differs from patient to patient (record 118 has a right bundle branch block), but after the template is subtracted a normal beat is close to zero everywhere and only the deviation of an ectopic beat from that patient's usual beat remains.

RR intervals are divided by the mean RR interval $\overline{RR}$ of the recording, $\tilde{r}_b = RR_b^{\mathrm{prev}} / \overline{RR}$. Neither step uses a label. The 150 s window comes from a training-patient diagnostic: S beats inside runs do not look premature against the ten preceding intervals, and only 51% of training S beats had a ratio below 0.9 that way, against 87% against the median of a ±150 s window. Ten independently trained runs are averaged, because single runs



(a) (b)

Figure 1: The system and its output. (a) The two pipelines stage by stage: left, the pre-registered system; right, the patient-context pipeline with the best test score (Section 5.3); the repository's full diagram also lists every alternative tested. (b) The output for one S beat of a validation patient: the rhythm strip with each beat's predicted class and the reference label, where the model looked, the probabilities, and how all 36 saved runs classified the beat (only 6 called it S).

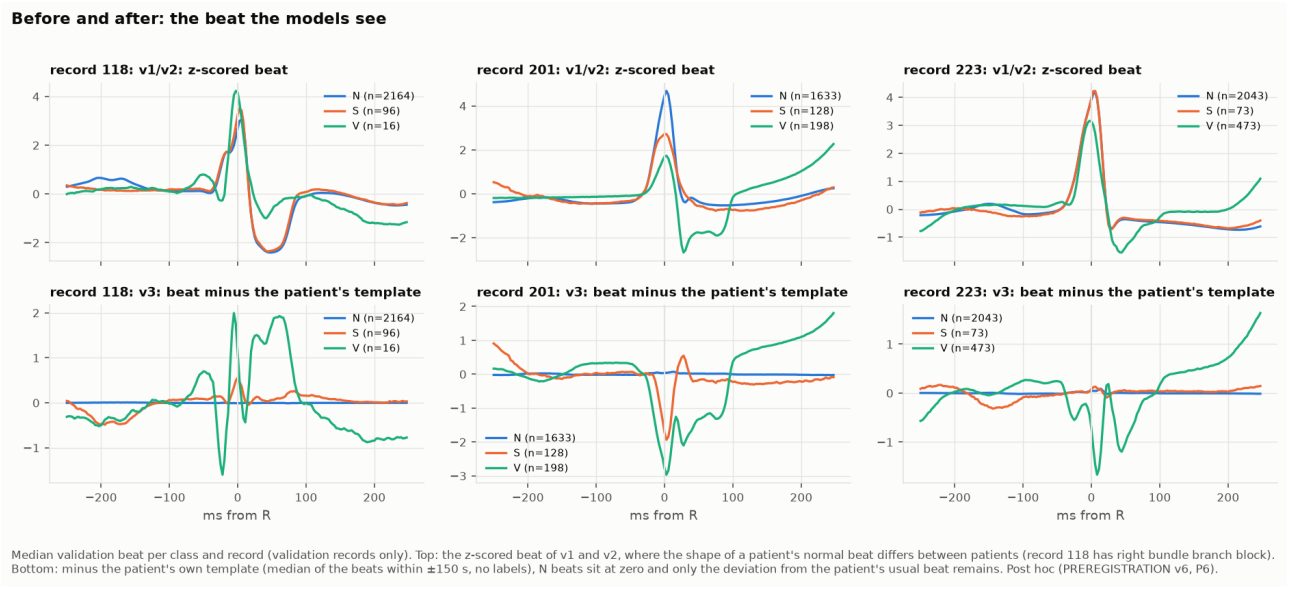


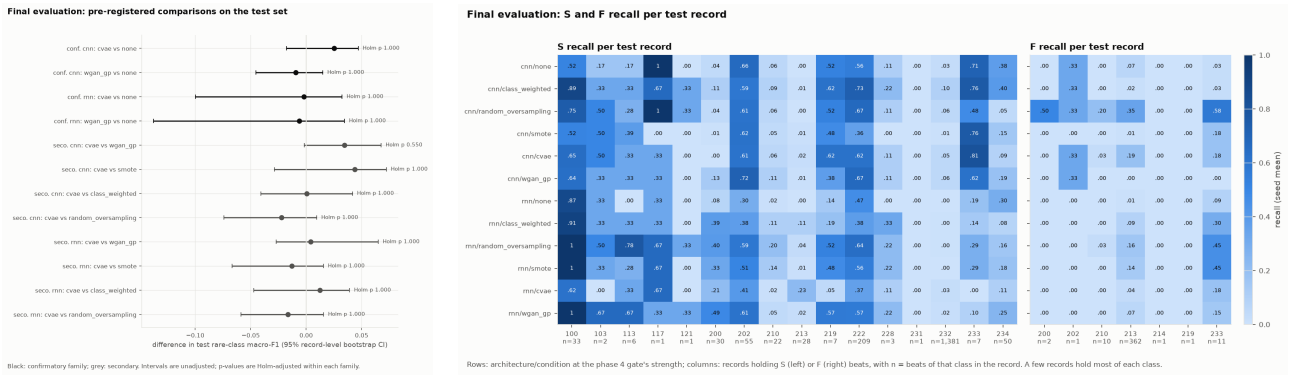
Figure 2: What the classifier sees, for three validation patients (median beat per class). Top: z-scored beats, whose shape differs between patients. Bottom: the same beats minus each patient's template (Equation 2); a normal beat is close to zero and an ectopic beat keeps only its deviation from the patient's usual beat.

of the same setting varied widely (0.451 to 0.610 on the test patients), $\bar{p}(c \mid b) = \frac{1}{10}\sum_{k=1}^{10} p_k(c \mid b)$, and a beat the ensemble calls normal becomes S when it is premature, unless an atrial-fibrillation detector fires:

$$\hat{y}_b = S \quad \text{if} \quad \arg\max_c \bar{p}(c \mid b) = N, \;\; \tilde{r}_b < 0.8 \;\; \text{and} \;\; a_b < 0.95, \tag{3}$$

and $\hat{y}_b = \arg\max_c \bar{p}(c \mid b)$ otherwise. The threshold 0.8 is Huang et al.'s [10], not tuned; a_b is the atrial-fibrillation probability of a logistic regression on five RR-irregularity measures, fitted on the 25 records of the MIT-BIH Atrial Fibrillation Database [7] (other subjects), with its threshold chosen on training-side data, never on the test records. Training also adds the patients of two further databases (St Petersburg INCART and the MIT-BIH Supraventricular Arrhythmia Database [7]), and the generators are retrained on patient-relative beats.

Intended setting. The template window is symmetric and includes the beat itself (one of several hundred beats



(a) (b)

Figure 3: The pre-registered comparison on the test patients. (a) Differences in rare-class macro-F1 with 95% intervals and Holm-adjusted p: black, the confirmatory family; grey, the secondary one. (b) S and F recall per test record for all twelve models: two records (232 for S, 213 for F) hold most rare beats, and every model misses most of them.

in the median), and the RR interval after the beat is an input, so the pipeline is designed for retrospective review of a recording, as in Holter analysis, with a latency of at least 150 s; it is not a real-time detector, and a past-only template was not tested.

4.3 Explanations and notes

Saliency is summarised as the share of attention in fixed P, QRS, ST and T windows around the R peak. A language model (Gemini) writes a short note from a JSON holding only the prediction, the probabilities, a low-confidence flag and the two most salient segments; a fact checker rejects any number, class, segment or finding not in that JSON and falls back to a template note. The checker guarantees consistency with the classifier, not clinical correctness: a wrong prediction is restated faithfully.

5 Results

Results at a glance (C: confirmatory; E: exploratory, repeated test reads; I: independent, untouched patients)

- C **Synthetic beats alone did not help:** best gain +0.025 (95% CI −0.018 to +0.047), no comparison significant after Holm's correction; **no quality metric predicted the gain**.
- E **Patient context:** rare-class macro-F1 from 0.113 to **0.587** [0.086, 0.702] on the same test patients, with fusion-beat F1 **0.527** and S F1 **0.647**.
- I **On 977 untouched patients**, the frozen pipeline's S F1 is **0.294** with its original thresholds, and **0.414** with its thresholds adapted on 388 separate Icentia11k patients; ten-run averaging helps there, the prematurity rule does not.
- E **Why synthesis fails:** 57–65% of synthetic S beats copy a single training patient, record 209.
- E **Patient overlap inflates scores almost tenfold: 0.837** on seen patients against **0.088** on unseen ones.

5.1 The pre-registered answer: synthetic beats did not help

Table 2 gives the single test read and Table 3 the registered hypotheses. For the CNN, no augmentation scored 0.088, the cVAE 0.113 and the WGAN-GP 0.079, against 0.113 for the class-weighted loss and 0.136 for random oversampling. The cVAE's gain, **+0.025** (95% CI −0.018 to +0.047), points the right way and nearly tripled the CNN's sensitivity for F beats (0.064 to **0.185**), but **no confirmatory comparison survived Holm's correction** (all adjusted $p = 1.000$), with or without record 202, whose patient also appears in training as record 201. Against the conventional remedies the cVAE was **level**: +0.001 against the class-weighted loss and −0.022 against random oversampling. For the BiLSTM all differences were smaller still (Figure 3a). On two other databases, St Petersburg INCART and the MIT-BIH Supraventricular Arrhythmia Database, generative augmentation was detectably better than no augmentation in 1 and worse in 1 of 8 comparisons. Normal and ventricular beats were detected at about the published level (CNN with the cVAE: N recall 0.938, V F1 0.893); S and F beats were not.

Where the errors are. Two test patients set most of the primary measure (Figure 3b): across the twelve models, S recall on record 232 ranged from 0.00 to 0.10, and F recall on record 213 from 0.00 to 0.35. The CNN without augmentation labelled 1,367 of the 1,836 S beats and 333 of the 388 F beats as normal, and

Architecture	Condition	Strength	Rare macro-F1 [95% CI]	N recall
cnn	none	–	0.088 [0.037, 0.194]	0.962
cnn	class weighted	0.5	0.113 [0.057, 0.207]	0.933
cnn	random oversampling	0.1	0.136 [0.061, 0.247]	0.844
cnn	smote	0.1	0.070 [0.025, 0.172]	0.985
cnn	cvae	0.25	0.113 [0.049, 0.217]	0.938
cnn	wgan gp	0.1	0.079 [0.033, 0.164]	0.955
rnn	none	–	0.075 [0.014, 0.232]	0.961
rnn	class weighted	0.75	0.061 [0.022, 0.168]	0.865
rnn	random oversampling	0.1	0.090 [0.028, 0.196]	0.883
rnn	smote	0.1	0.086 [0.025, 0.206]	0.894
rnn	cvae	0.1	0.073 [0.035, 0.157]	0.926
rnn	wgan gp	0.5	0.069 [0.027, 0.156]	0.857

Table 2: The pre-registered comparison: the 22 test patients, read once. Rare-class macro-F1 with 95% intervals from a bootstrap over patients, and the recall of normal beats; the strength s is the fraction of the N count chosen on validation.

Hypothesis	Statement	Verdict	Evidence
H1	confirmatory: a generative condition improves test rare-class macro-F1 over none (Holm, 4 tests)	not supported	no confirmatory comparison significant
H1b	exploratory: gains larger for F than for S	mixed	F gain larger than S gain in 2 of 4 generative conditions
H1c	secondary: generative beats SMOTE, random oversampling and class-weighted loss	not supported	0 of 6 comparisons of the best generative condition with a baseline significant
H2	exploratory: intra-patient evaluation inflates the baseline and the gains	not consistent	inflation of the baseline +0.748; larger intra-patient gain for 0 of 2 generators
H3	exploratory: a quality metric predicts the downstream gain	not supported	0 of 5 metrics with an interval excluding zero
H4	exploratory: augmented models are at least as faithful as the baseline	mixed	augmented faithfulness at least the baseline's in 1 of 4 model pairs

Table 3: The registered hypotheses, their verdicts (the rules applied as written) and the evidence.

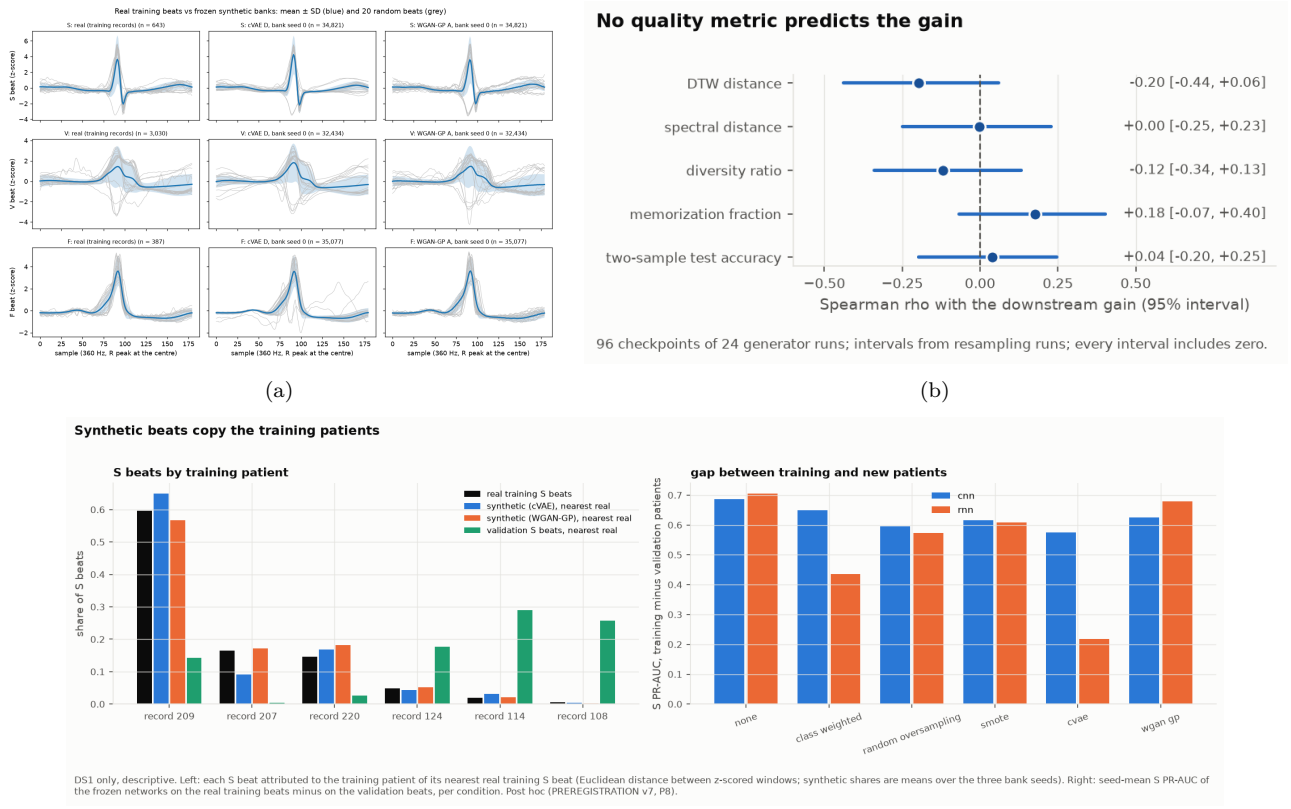
the S beats it did find were **more premature** than those it missed (median ratio of the previous RR interval to the local mean **0.57 against 0.72**). The model found the S beats that broke the rhythm most and missed those close to it, which pointed the follow-up analyses to judging prematurity against each patient's own rhythm.

5.2 Why the synthetic beats did not transfer

Before use, the quality metrics were checked on deliberately degraded beats: dynamic time warping and a spectral distance ranked them correctly, a Fréchet-distance analogue and train-on-synthetic-test-on-real did not. Five metrics covering fidelity, diversity, memorization and a classifier two-sample test were then compared with the downstream gain over 96 checkpoints of 24 generator runs; **none predicted it** (Spearman ρ from -0.20 to $+0.18$, every interval including zero). The beats themselves show why. **Between 57% and 65% of the synthetic S beats lie nearest to a single training patient**, record 209, just as 60% of the real training S beats do, while the validation patients' S beats lie nearest to training patients who contribute **only 1–2%** of them. The generators reproduce the training distribution faithfully, and that distribution is dominated by one patient; a new patient's rare beats are not in it (Figure 4). Two alternatives did not change this: unscreened selection policies lost to the DTW screen in the policy pilot on validation, and patient transfer, which grafts each rare beat's deviation onto another training patient's template, was not detectably better than no augmentation on the test patients (BiLSTM 0.075 to 0.105–0.109; CNN 0.096 to 0.089–0.116). These conclusions hold for the generators, the 17 training patients and the selection policies tested here, not for generative augmentation in general.

5.3 Patient context: an exploratory pipeline

Figure 5 traces the follow-up analyses on the test patients. These are exploratory: each analysis was registered before it ran, but each chose its next step after seeing earlier test scores, so the scores below are optimistic. Patient-relative beats raised the CNN without augmentation from 0.088 to **0.371** (P7), **the largest single step**. Generators retrained on patient-relative beats lifted the cVAE condition to **0.368** (P12). Normalising RR intervals by the recording's mean with Huang et al.'s rule on top raised it to **0.523** (P13); the atrial-fibrillation veto raised every condition by 0.018 to 0.034, each 95% interval above zero (P16); and averaging ten runs gave **0.587** [0.086, 0.702], with S F1 **0.647** and F F1 **0.527** (P18). Within this pipeline the cVAE scored highest, 0.046 to 0.271 above the other conditions, mostly through fusion beats, but none of these differences is detectable



(a) (b) (c)

Figure 4: Why the synthetic beats did not transfer. (a) Real training beats beside the frozen cVAE and WGAN-GP banks (mean ± SD and 20 random beats): the synthetic beats look right. (b) The Spearman correlation of each quality metric with the downstream gain over 96 checkpoints of 24 generator runs: every 95% interval includes zero. (c) Each S beat attributed to its nearest training patient: synthetic beats follow the real training mix, dominated by record 209, while the validation patients' S beats are nearest to patients the training set barely holds.

	S F1	V F1	F F1	Mean of S and F
Test patients unseen in training (DS2)				
de Chazal et al. 2004 [5]	0.511	0.797	–	–
Huang et al. 2014 [10]	0.577	0.924	–	–
Zhu et al. 2022 [24]	0.686	0.960	0.477	0.582
Mousavi et al. 2019† [15]	0.907	0.997	–	–
Ours, RR-only linear discriminant, same split (P1)				0.022
Ours, pre-registered (CNN with the cVAE)	0.173	0.893	0.054	0.113
Ours, patient-context pipeline‡	0.647	0.936	**0.527**	**0.587**
Test patients also seen in training (beats split at random)				
Kwak and Jung 2024 [11]	0.947	0.985	0.903	0.925
Ours, the same CNN (no augmentation)				0.837

Table 4: Beside published results, for context only: the published rows differ from ours in training data, preprocessing and test reads, so this is not a controlled comparison; the controlled comparisons are Table 2 and the RR-only discriminant row, run on exactly our split. F1 is computed from the reported sensitivity and positive predictivity where a paper gives no F1. Bold: the best inter-patient value in the column. †Far above every other inter-patient report; its protocol could not be checked here. ‡Post hoc, and trained with patients of two further databases as well as DS1.

with 22 test patients (Holm p from 0.107 to 1.000): synthesis on top of context is a promising point estimate, not a finding.

Table 4 sets the pipeline beside published results, for context. It is **above the two classic methods on both classes they report**, S F1 0.647 against 0.511 and 0.577 and V F1 0.936 against 0.797 and 0.924, and **above Zhu et al.'s recent method on fusion beats (0.527 against 0.477) and on the mean of S and F (0.587 against 0.582)**, while Zhu et al. remain ahead on S (0.686) and V (0.960) beats. Three qualifications apply: our pipeline was trained with additional patients from two further databases, it was built while reading the same test patients more than once, and the published numbers come from other preprocessing and code. On sensitivity and precision, the pipeline finds 0.914 of the test S beats with a precision of 0.500, against 0.152

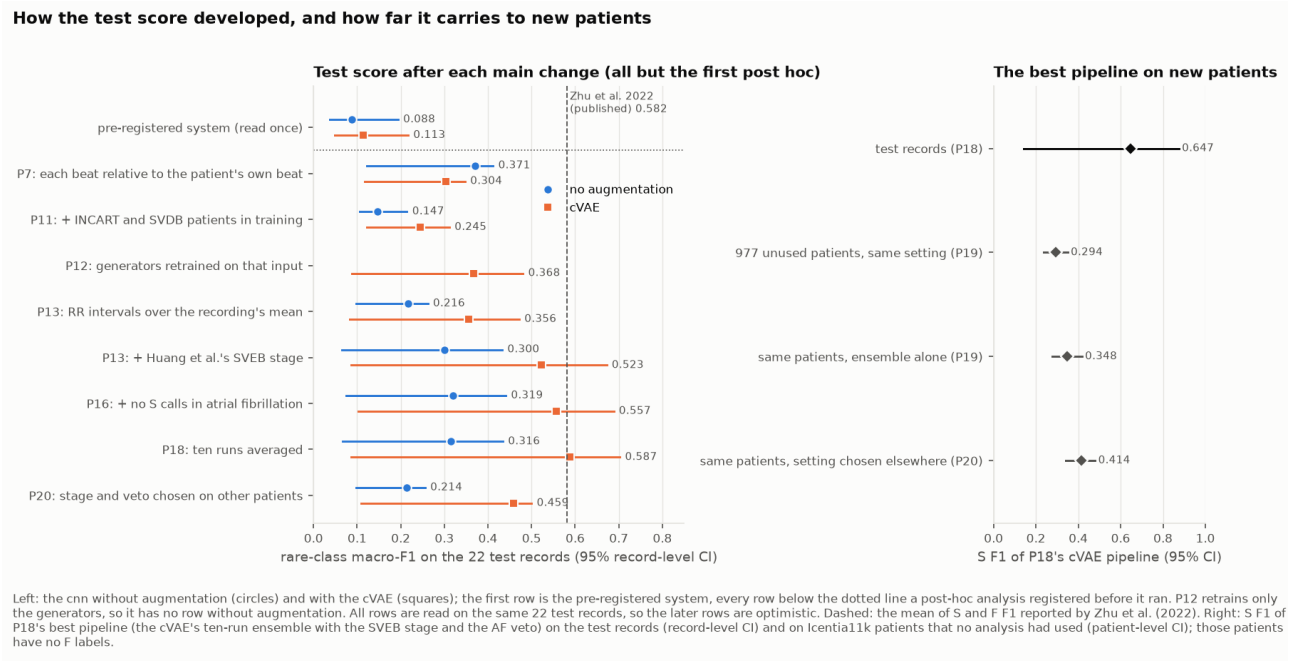


Figure 5: The test score after each main change, without augmentation (circles) and with the cVAE (squares); every row below the dotted line is a follow-up analysis; dashed, Zhu et al.'s published mean of S and F F1. Right: the best pipeline's S F1 on the test patients and on unseen Icentia11k patients.

Step	Test patients (E)	Evidence without the test set (V: validation; I: untouched)
Patient-relative beats (P7)	CNN, no augmentation: 0.088 → **0.371**	V: S F1 CNN 0.350 → 0.472; BiLSTM 0.306 → 0.181
MIT-BIH only vs. + INCART, SVDB (P11)	no augmentation 0.371 → 0.147; class-weighted 0.407	–
RR over the recording mean, Huang's rule (P13)	cVAE 0.356 → **0.523**	I: S Se 0.303 → 0.629 but S PPV 0.408 → 0.157
AF veto (P16)	+0.018 to +0.034, every CI above zero; S PPV 0.415 → 0.496	Threshold set on training-side data (S F1 0.490 → 0.575); V: rule with veto 0.753 vs. CNN alone 0.755
Ten runs averaged (P18, P19)	cVAE 0.557 → **0.587** (not detectable)	I: S F1 +0.006 to +0.016 in 5 of 6 conditions, Holm p 0.001
Thresholds chosen on other patients (P20)	cVAE 0.587 → 0.459	I: S F1 0.294 → **0.414**, Holm p 0.001

Table 5: What each step of the patient-context pipeline contributed, on the test patients (exploratory) and where available without the test set. Rare-class macro-F1 unless marked; S Se and S PPV are the sensitivity and positive predictivity for S beats.

and 0.201 for the pre-registered CNN.

What each step contributed. The changes were added one after another, so their effects are not separable on the test patients alone. Table 5 sets each step's test effect beside the evidence that does not depend on the test set. Patient-relative beats helped the CNN on validation patients but hurt the BiLSTM; adding INCART and SVDB patients lowered the score without augmentation, so the extra data do not explain the gain; ten-run averaging and thresholds chosen on other patients were confirmed on untouched patients; the prematurity rule raised S sensitivity everywhere but, on untouched patients, cut precision more than it added recall. Other ideas tested in the same pipeline (a diffusion model, a physics prior from McSharry et al.'s simulator [13] as in SimGANs [6], and a ResNet1D with 6.7 times the CNN's parameters) were not detectably better than the cVAE (Holm p 0.24 to 0.62).

5.4 Independent check on untouched patients

To discount those repeated reads, the frozen pipeline was run on 977 patients of the Icentia11k database [23] that no analysis had used, with nothing fitted on them. Icentia11k differs from MIT-BIH in device (one chest lead at 250 Hz), annotation (technician-labelled N, S, V and Q beats) and population, and has no F label, so fusion beats could not be checked. Its S F1 was **0.294** [0.237, 0.350] with its own thresholds, and **0.414** when the two thresholds were chosen on 388 other Icentia11k patients; on the test records that choice gave 0.459,

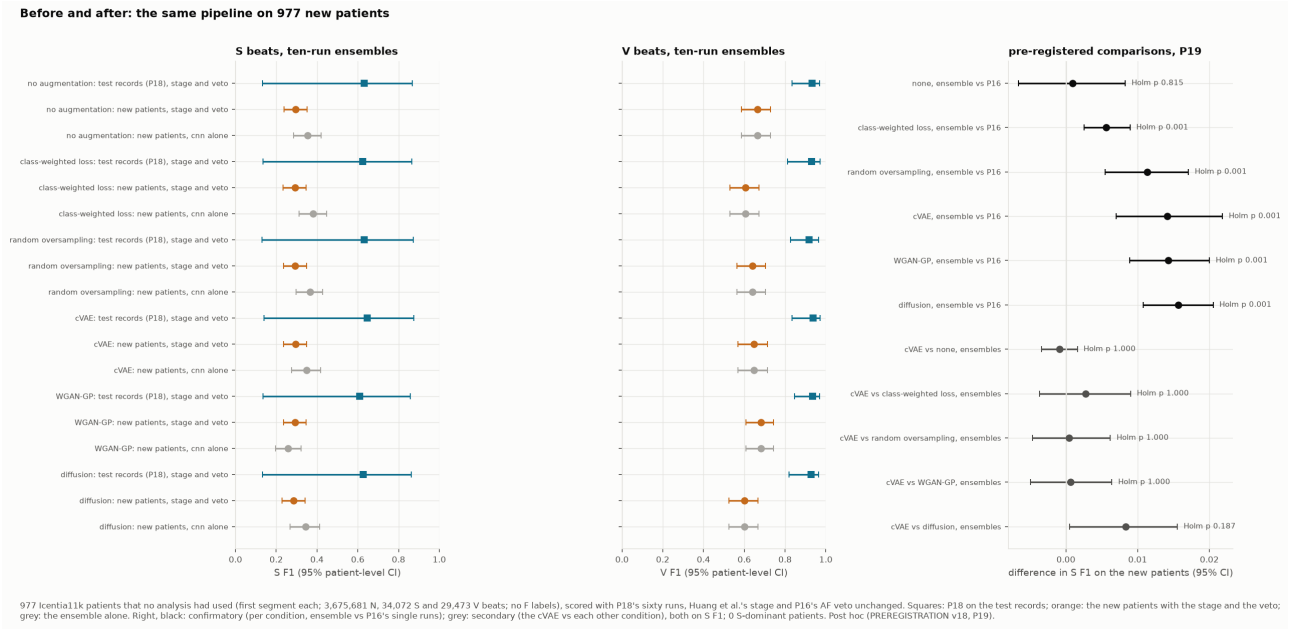
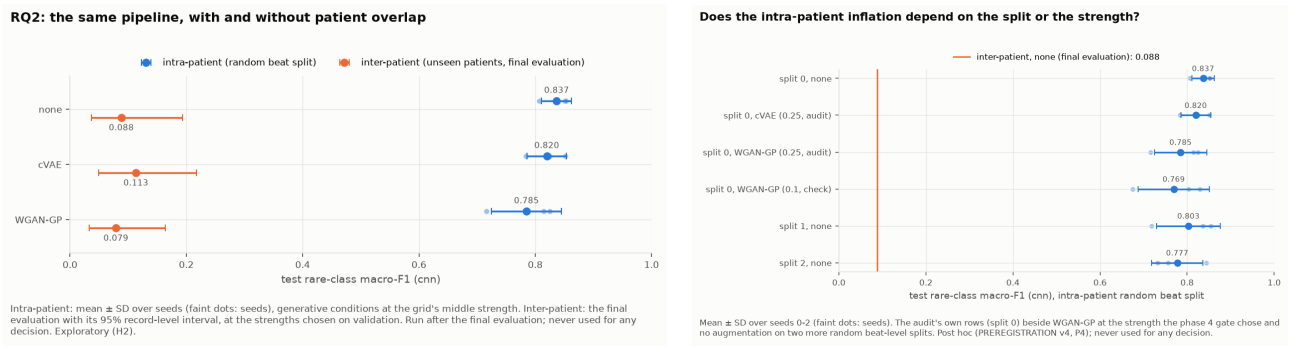


Figure 6: The frozen pipeline on 977 unseen Icentia11k patients. Left and middle: S and V F1 of each condition's ten-run ensemble on the test records (squares), on the unseen patients with the rule and the veto (orange) and without them (grey). Right: the registered comparisons on the unseen patients, with Holm-adjusted p.



(a) (b)

Figure 7: Patient overlap. (a) The same pipeline on held-out beats of the training patients (blue) and on unseen patients (orange), with and without generative augmentation. (b) The random-split score on two further splits and at a second WGAN-GP strength, against the unseen-patient score of 0.088 (line).

because the test record that holds most S beats resembles no tuning patient. On these patients the ensemble alone did better than with the rule (cVAE S F1 0.348): Huang et al.'s threshold, set on MIT-BIH, doubled S sensitivity (0.303 to 0.629) but cut positive predictivity (0.408 to 0.157), and V F1 fell from 0.936 on the test patients to 0.648 (Figure 6). **Patient context therefore transfers, but less than the test score suggests**, and the cVAE was level with the other conditions there (differences −0.001 to +0.008, none detectable).

5.5 Patient overlap inflates the score almost tenfold

After the final evaluation, the same CNN was trained and tested on a random beat split of all 44 records, so that beats of every patient were in both training and test. Without augmentation it reached **0.837** ± 0.026 (Figure 7a), and 0.803 and 0.777 on two further random splits (Figure 7b), against **0.088** on unseen patients. The gap is not a property of one condition: the cVAE reached 0.820 and the WGAN-GP 0.785 under the random split. Generative augmentation helped under neither protocol (with patients seen in training: cVAE −0.017, WGAN-GP −0.052). A model that has seen a patient's beats largely recognises that patient; random-split results above 0.9 therefore belong beside our own 0.837, not beside inter-patient results (Table 4), and every other number in this report is measured on patients unseen in training.

5.6 A prediction a clinician can check: the demonstrator

This part is the project's engineering demonstrator of generative AI in the loop, not a validated clinical explanation system. Of the four saliency methods, the registered rule chose Grad-CAM for the CNN, the most faithful

(deleting its most salient samples hurt the model most), and occlusion for the BiLSTM. Faithfulness here means the saliency reflects what the model used, not that the model is clinically right. Of four note generators tried on 50 beats, **all made zero unsupported claims**; the rule chose Gemini with worked examples and a fixed output schema, which covered every required fact in 2.6 s per note. the notes were rated by team members, not clinicians, and 50 beats is too few for strong reliability claims. Figure 1b shows the demonstrator for one S beat: the offline analysis by our models, which a review written by Gemini from those results then restates, claim by claim checked.

6 Discussion

Context before synthesis. With 17 training patients, the bottleneck for unseen patients is not the number of rare beats but the number of patients they come from. A generator trained on that set multiplies its dominant patient; what transferred instead was to read each beat against its own patient, as a cardiologist reads a rhythm strip: is the beat early for this heart, does it differ from this patient's normal beat, is the rhythm fibrillating? Once beats were expressed that way, the cVAE gave the best point estimate, which suggests the order in which the two ideas should be combined.

The protocol decides the score. The same network scores 0.837 or 0.088 depending only on whether the test patients were seen in training, so results on random beat splits cannot be ranked against inter-patient ones.

Reproducibility. Every number in this report is written by code from logged runs, each with its configuration, seed, commit and environment. The registered rules (21 versions: versions 1 to 3 for the pre-registered study and 4 to 21 for the 23 follow-up analyses, some versions registering more than one), every decision and every read of the test records are kept in the project repository (`https://github.com/akashsaha477/Gen_Ai_Project`), the data are public on PhysioNet, and the test suite, including a dress rehearsal that runs every phase, gate and evaluation in order on fake records, passes on a fresh copy of the repository.

Limitations. *Evidence:* the patient-context pipeline is exploratory; only the untouched Icentia11k patients test it cleanly, and there it is weaker (S F1 0.294 to 0.414) and untested on fusion beats. *Sample:* the rare test beats sit in two patients, so intervals are wide (0.086 to 0.702) and one record can move the primary measure; per-patient S and F recall is shown in Figure 3b, while a per-patient distribution for the follow-up pipeline would need a further test read, which we did not make. *Selection:* models were selected on validation S F1 because validation holds only 27 F beats, so F performance was never optimised directly. *Metrics:* beyond F1 we report sensitivity, precision and N recall, but not specificity, calibration, AUROC or false alarms per hour. *Setting:* the pipeline uses the reference R peaks, one lead (MLII), a retrospective 150 s template and ten models at inference (each trained in 38–74 s on a T4 GPU); an automatic R-peak detector, other leads, causal templates and inference cost were not evaluated. The notes restate the classifier and do not diagnose.

7 Conclusion

The central finding of this project is where the difficulty lies: rare arrhythmias are hard to detect in new patients because every heart is different, and the promising way through that patient-level shift is patient-relative context, not synthetic data. We reached it by asking, under rules fixed before the test data was read, whether synthetic heartbeats can help, and by following the answer.

1. **Patient context is the most promising direction (exploratory, partly confirmed).** Reading each beat against its own patient and averaging ten runs raised the rare-class macro-F1 on the test patients from 0.113 to **0.587**; on 977 untouched patients the frozen pipeline reached an S F1 of **0.294** with its original thresholds, and **0.414** after adapting its two thresholds on 388 separate Icentia11k patients; ensembling and that adaptation held up there, the prematurity rule did not.
2. **The tested synthetic beats did not help (confirmatory).** A conditional VAE and a WGAN-GP gave no detectable gain over no augmentation or simple balancing (+0.025 at best), because they faithfully reproduce the few training patients they learn from; this negative result is what pointed us to patient context.
3. **Evaluation must keep patients apart.** The same network scores **0.837** on patients it has seen and **0.088** on new ones.

The decisive next experiment needs no new idea, only new patients: freeze the patient-context pipeline, ablate its parts one at a time, compare causal with symmetric templates, and test once on an untouched cohort that carries fusion-beat labels.

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